

DS360 – Project Phase 2

WAN 1 Configuration – Detailed

1. Project Objective

The objective of this part of the DS360 project is to build and configure WAN 1 in Cisco Packet Tracer. WAN 1 contains two local area networks (LAN 1 and LAN 2) connected through Router0. The work includes creating the topology, connecting PCs to switches and switches to the router, assigning IPv4 addresses, configuring the two Router0 LAN interfaces, and verifying that all interfaces and local connections are working correctly.

2. Required WAN 1 Devices and Structure

- 1 × Cisco 1941 Router: Router0
- 2 × Cisco 2960-24TT Switches: Switch0 and Switch1
- 6 × PCs: PC0, PC1, PC2, PC3, PC4, and PC5
- LAN 1 connects PC0–PC2 to Switch0, then Switch0 to Router0 GigabitEthernet0/0.
- LAN 2 connects PC3–PC5 to Switch1, then Switch1 to Router0 GigabitEthernet0/1.

3. Proposed WAN 1 Topology

The topology should be arranged as follows:

```

PC0, PC1, PC2 → Switch0 → Router0 ← Switch1 ← PC3, PC4, PC5
LAN 1: 172.16.10.0/24                                LAN 2: 172.16.20.0/24

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Use Copper Straight-Through cables for all PC-to-switch and switch-to-router links. The assignment requires verifying that the connected ports are active before moving to IP configuration.

4. IPv4 Addressing Plan

The assignment explicitly defines the two /24 networks and the Router0 gateway addresses. It does not specify the exact host address for each PC. Therefore, the PC addresses below are a suggested valid implementation inside the required networks. Any unique host addresses in the correct subnet would also work.

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway	Status
Router0	G0/0	172.16.10.1	255.255.255.0	N/A	Required
PC0	FastEthernet0	172.16.10.2	255.255.255.0	172.16.10.1	Suggested
PC1	FastEthernet0	172.16.10.3	255.255.255.0	172.16.10.1	Suggested
PC2	FastEthernet0	172.16.10.4	255.255.255.0	172.16.10.1	Suggested
Router0	G0/1	172.16.20.1	255.255.255.0	N/A	Required
PC3	FastEthernet0	172.16.20.2	255.255.255.0	172.16.20.1	Suggested
PC4	FastEthernet0	172.16.20.3	255.255.255.0	172.16.20.1	Suggested
PC5	FastEthernet0	172.16.20.4	255.255.255.0	172.16.20.1	Suggested

5. Detailed Packet Tracer Implementation

5.1 Day 1 – 03 August 2026: Create the WAN 1 Topology

1. Open Cisco Packet Tracer and create a new project file.
 2. From Network Devices > Routers, drag one Cisco 1941 router into the workspace and keep the default name Router0.
 3. From Network Devices > Switches, add two Cisco 2960-24TT switches. Keep their names Switch0 and Switch1.
 4. From End Devices, add six PCs and ensure they are labeled PC0 through PC5.
 5. Place PC0, PC1, and PC2 near Switch0 to represent LAN 1.
 6. Place PC3, PC4, and PC5 near Switch1 to represent LAN 2.
 7. Place Router0 between the two switches so the two LANs clearly connect to the same router.
 8. Save the Packet Tracer file after the topology is created.
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5.2 Day 2 – 04 August 2026: Connect the LAN Devices

9. Select Connections and choose Copper Straight-Through.
 10. Connect PC0 FastEthernet0 to an available FastEthernet port on Switch0 (for example Fa0/1).
 11. Connect PC1 FastEthernet0 to Switch0 Fa0/2 and PC2 FastEthernet0 to Switch0 Fa0/3.
 12. Connect Switch0 to Router0 GigabitEthernet0/0. A switch Gigabit port such as Gi0/1 is a convenient choice.
 13. Connect PC3 FastEthernet0 to Switch1 Fa0/1, PC4 to Switch1 Fa0/2, and PC5 to Switch1 Fa0/3.
 14. Connect Switch1 to Router0 GigabitEthernet0/1 using a switch Gigabit port such as Gi0/1.
 15. Wait for link indicators to become green. If a link remains red, check the cable type and router interface status.
 16. Save the Packet Tracer file after confirming the physical connections.
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5.3 Day 3 – 05 August 2026: Configure IPv4 Addressing

Configure the PCs first. For each PC, open Desktop > IP Configuration and enter the address, mask, and gateway from the addressing table above.

- PC0: IP 172.16.10.2, Mask 255.255.255.0, Gateway 172.16.10.1
- PC1: IP 172.16.10.3, Mask 255.255.255.0, Gateway 172.16.10.1
- PC2: IP 172.16.10.4, Mask 255.255.255.0, Gateway 172.16.10.1
- PC3: IP 172.16.20.2, Mask 255.255.255.0, Gateway 172.16.20.1
- PC4: IP 172.16.20.3, Mask 255.255.255.0, Gateway 172.16.20.1
- PC5: IP 172.16.20.4, Mask 255.255.255.0, Gateway 172.16.20.1

6. Router0 Configuration Commands

Open Router0 > CLI. If the initial configuration dialog appears, answer No. Then enter the following commands:

```
enable
configure terminal
interface gigabitEthernet0/0
ip address 172.16.10.1 255.255.255.0
```

```

no shutdown
exit
interface gigabitEthernet0/1
ip address 172.16.20.1 255.255.255.0
no shutdown
end
show ip interface brief

```

Expected result: GigabitEthernet0/0 should display IP address 172.16.10.1 and GigabitEthernet0/1 should display 172.16.20.1. After the links initialize, both interfaces should be up/up. The no shutdown command is essential because router interfaces are commonly administratively down until enabled.

7. Verification and Testing

Perform the following checks before taking evidence screenshots:

- On Router0, run show ip interface brief and confirm G0/0 and G0/1 use the correct IP addresses and are up/up.
- From PC0, open Desktop > Command Prompt and ping its gateway: ping 172.16.10.1.
- From PC3, ping its gateway: ping 172.16.20.1.
- Test within LAN 1, for example from PC0: ping 172.16.10.3 and ping 172.16.10.4.
- Test within LAN 2, for example from PC3: ping 172.16.20.3 and ping 172.16.20.4.
- Test communication between the two LANs through Router0, for example from PC0: ping 172.16.20.2. Because both networks are directly connected to Router0, no additional static route is required for this WAN 1 local inter-LAN test.
- If the first ping times out once, repeat it. Packet Tracer may need to learn ARP/MAC information before later echo replies succeed.

8. Troubleshooting Guide

Problem	Likely Cause	Fix
Router port is red/down	Interface is administratively shut down	Enter interface configuration mode and use no shutdown.
PC cannot ping its gateway	Wrong IP, mask, gateway, or cable	Recheck Desktop > IP Configuration and the switch/router connection.
PCs in same LAN cannot ping	Duplicate or wrong host IP / physical link issue	Make each PC address unique and confirm green switch links.
PC in LAN 1 cannot ping LAN 2	Gateway missing/wrong or Router0 interface down	Verify PC gateways and show ip interface brief on Router0.
show ip interface brief shows down/down	Physical link or remote port problem	Check straight-through cable and ensure the switch side is connected.

9. Evidence to Upload

The assignment requires real evidence from the actual Packet Tracer work. Do not fabricate or backdate screenshots or commit history.

- Screenshot of the complete WAN 1 topology showing Router0, Switch0, Switch1, and PC0–PC5.
- Screenshot showing the LAN 1 and LAN 2 physical connections.
- Screenshot(s) of the end-device IPv4 configuration for LAN 1 and LAN 2.
- Screenshot of Router0 CLI showing the configured LAN interfaces and/or the show ip interface brief output.

10. GitHub Submission Notes

Upload the actual Packet Tracer project file and the real evidence that corresponds to the work completed. Commit messages should describe genuine changes. The assignment specifically warns against creating backdated or fabricated project history.

- 03 August: Create WAN1 network topology
- 04 August: Add WAN1 LAN connections
- 05 August: Configure WAN1 IP addressing

11. Final Work Summary

WAN 1 was designed with two LANs connected to Router0. LAN 1 uses network 172.16.10.0/24 with Router0 G0/0 as gateway 172.16.10.1, while LAN 2 uses network 172.16.20.0/24 with Router0 G0/1 as gateway 172.16.20.1. Six PCs are connected through two 2960 switches using Copper Straight-Through cables. After IPv4 configuration and enabling both router interfaces, the network should support successful communication within each LAN and between LAN 1 and LAN 2 through Router0. Final verification should be demonstrated using real Packet Tracer screenshots and ping tests.

12. Submission Checklist

- ☐ Packet Tracer topology contains Router0, Switch0, Switch1, and PC0–PC5.
- ☐ LAN 1 uses 172.16.10.0/24 and gateway 172.16.10.1.
- ☐ LAN 2 uses 172.16.20.0/24 and gateway 172.16.20.1.
- ☐ All end devices use subnet mask 255.255.255.0 and the correct gateway.
- ☐ Router0 G0/0 and G0/1 are configured and enabled with no shutdown.
- ☐ Physical links are active/green.
- ☐ Ping tests are successful.
- ☐ Required screenshots are captured from the real project.
- ☐ Actual Packet Tracer/project evidence is uploaded to GitHub with truthful commit history.